



Faculty of Technology
University of Sri Jayewardenepura

ITC4182 Professional Practices in ICT Assignment 2

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1. Introduction to ICT Professional Organizations

Professional organizations play an important role in the Information and Communication Technology (ICT) industry. These organizations bring together ICT professionals, students, researchers, and companies to work towards common goals. They help improve the quality of ICT work, set standards that everyone follows, and make sure ICT professionals behave ethically.

ICT professional organizations do several important things. First, they create and maintain standards that guide the development and use of technology. Second, they provide training and learning opportunities so professionals can keep their skills up to date in this fast changing field. Third, they connect people working in similar areas, helping them share knowledge and find job opportunities. Fourth, they give professionals a voice when talking to governments and making policies that affect the industry.

2. Overview of Selected Organizations

2.1. IEEE (Institute of Electrical and Electronics Engineers)

IEEE is one of the largest IT related organizations in the world. This was established in 1963. It has over 500,000 members across 160 countries. IEEE operates through 10 global regions. It has 39 technical societies. These societies cover various technology areas. The main areas include “Electrical Engineering, Computer Science, Telecommunications, Artificial Intelligence, Cybersecurity, Robotics, and Biomedical Engineering”

2.2. ACM (Association for Computing Machinery)

The ACM organization was founded in 1947. right after World War II, when computers were just beginning. Now, it has around 110,000 members worldwide and is one of the most important organizations for computing professionals. They have a motto, which is “Advancing Computing as a Science & Profession.”

2.3. CSSL (Computer Society of Sri Lanka)

CSSL (Computer Society of Sri Lanka) was established in 1976. Nowadays, CSSL is the main professional body representing ICT professionals in Sri Lanka. It was incorporated in 1986 and has been working for nearly 50 years to improve ICT standards and professionalism in our country.

2.4. SLASSCOM (Sri Lanka Association of Software and Service Companies)

SLASSCOM (Sri Lanka Association of Software and Service Companies) was founded in 2008. SLASSCOM (Sri Lanka Association of Software and Service Companies) primarily represents companies in the software and IT services sector. Currently, it has over 400 member companies employing more than 115,000 people.

3. Comparative Analysis

3.1 Focus and Objectives

3.1.1 IEEE (Institute of Electrical and Electronics Engineers)

IEEE focuses on advancing technology across all engineering and technical fields. IEEE improves technology in all engineering and technical areas. Their Objectives are to encourage new ideas, set industry standards, support an ethical technology environment, and help professionals in their careers. IEEE covers a wide range of areas, including electrical engineering, electronics, computer science, telecommunications, and new technologies. They operate worldwide and want to be the top trusted source for technical information and standards.

3.1.2 ACM (Association for Computing Machinery)

ACM's focus is specifically on computing and computer science. Their main objectives are to advance computing as both a science and a profession. And support computing research, and ensure ethical practices in the field. Another major objective is educating policymakers about computing. ACM works to help people understand both the benefits and potential risks of computing technology, including important topics like artificial intelligence, data privacy, and cybersecurity. They promote the development and use of environmentally sustainable computing technologies and encourage computing professionals to adopt responsible practices that minimize environmental impact.

3.1.3 CSSL (Computer Society of Sri Lanka)

CSSL's primary focus is on promoting ICT professionalism specifically within Sri Lanka. Their objectives are to maintain professional standards among Sri Lankan ICT practitioners, ensuring quality ICT education through degree accreditation, representing Sri Lankan ICT professionals internationally, and fostering research and knowledge sharing. CSSL (Computer Society of Sri Lanka) works to connect the local ICT community with international standards and practices while addressing the specific needs of the Sri Lankan market.

3.1.4 SLASSCOM (Sri Lanka Association of Software and Service Companies)

SLASSCOM (Sri Lanka Association of Software and Service Companies) has a more business-oriented focus compared to the other three organizations. Their main objectives are to grow Sri Lanka's IT export industry, create employment opportunities, develop a skilled workforce, create startups, and innovate. And also position Sri Lanka as a global technology hub. Rather than focusing on individual professional development, SLASSCOM concentrates on helping companies grow, access international markets, and contribute to Sri Lanka's economy.

3.2 Key Activities and Services

3.2.1 IEEE (Institute of Electrical and Electronics Engineers)

IEEE organizes over 2,000 conferences and events annually worldwide, covering every aspect of technology. These conferences allow professionals and researchers to present their work, learn about new developments. IEEE publishes more than 200 journals, magazines, and conference proceedings, making it one of the most cited sources in technical literature. Their

publications include the prestigious IEEE Transactions series and popular magazines like IEEE Spectrum.

3.2.2 ACM (Association for Computing Machinery)

ACM's primary activity is to publish high-quality computing research. They publish over 50 journals and magazines, including the famous Communications of the ACM, which has been running since 1958. The ACM Digital Library contains over 2 million pages of research papers and is considered the most comprehensive collection of computing literature in the world.

3.2.3 CSSL (Computer Society of Sri Lanka)

CSSL (Computer Society of Sri Lanka) "Flagship event" is the National Information Technology Conference (NITC), which has been running annually since 1981. This is Sri Lanka's premier ICT conference, bringing together over 1,000 ICT professionals, featuring more than 30 local and international speakers, and covering the latest technology trends. The conference serves as a platform for knowledge sharing and networking within the Sri Lankan ICT community.

3.2.4 SLASSCOM (Sri Lanka Association of Software and Service Companies)

SLASSCOM (Sri Lanka Association of Software and Service Companies) focuses heavily on helping Sri Lankan IT companies grow and access international markets. They provide export marketing training and coaching to help companies compete globally. One of SLASSCOM's major initiatives is workforce development. They've identified a significant gap between university education and industry requirements, so they run the Graduate to Corporate (G2C) program to improve the professionalism of fresh graduates

3.3 Code of Conduct and Ethics

3.3.1 IEEE (Institute of Electrical and Electronics Engineers)

IEEE's Code of Ethics, updated most recently in 2020. The key principles include prioritizing public safety, health, and welfare, protecting privacy, complying with ethical design practices, and disclosing any dangers to the public or environment. Members must avoid conflicts of interest, reject bribery, be honest in technical work, acknowledge mistakes, and maintain their technical competence.

3.3.2 ACM (Association for Computing Machinery)

It covers general ethical principles like contributing to society, avoiding harm, being honest, respecting privacy, and not discriminating. Leaders must ensure their organizations follow ethical principles and create opportunities for professional growth. ACM has a Committee on Professional Ethics (COPE) and an "Ask an Ethicist" service for members facing ethical dilemmas.

3.3.3 CSSL (Computer Society of Sri Lanka)

CSSL promotes high professional standards among Sri Lankan ICT professionals through its membership requirements and activities. They work with international bodies like IFIP and SEARCC, following their ethical guidelines. They introduce professional ethics to students early in their careers.

3.3.4 SLASSCOM (Sri Lanka Association of Software and Service Companies)

SLASSCOM, being a company-focused organization, emphasizes business ethics and corporate responsibility rather than individual professional codes.

3.4 Membership Structure

3.4.1 IEEE (Institute of Electrical and Electronics Engineers)

IEEE offers several membership grades for people at different career stages. Full professional membership costs \$218 per year and is open to anyone working in technology fields.

3.4.2 ACM (Association for Computing Machinery)

ACM has a simpler membership structure. Student membership costs around \$19 per year and is for those enrolled in academic programs.

3.4.3 CSSL (Computer Society of Sri Lanka)

CSSL has different memberships Associate membership is for those starting their ICT careers. Full membership (MCS - Member of Computer Society) is for established ICT professionals. Honorary Fellow membership is awarded to distinguished individuals who have made exceptional contributions. Corporate membership is available for companies.

3.4.4 SLASSCOM (Sri Lanka Association of Software and Service Companies)

They have different membership categories based on company size, type, and revenue.

4. Benefits for ICT Students and Graduates

4.1. Meeting people and building connections

IEEE and ACM provide global networking through thousands of student chapters worldwide.

4.2. Access to learning materials and research

IEEE offers 5 million technical documents, while ACM provides 2 million research papers (free from 2026).

4.3. Career growth opportunities

Participating in competitions like IEEE Xtreme or ACM ICPC builds practical skills.

4.4. Getting recognized in the industry

Organizations provide certificates for participation and achievements.

4.5. Finding mentors

IEEE and ACM have formal mentorship programs connecting students with experienced professionals globally.

5. Critical Evaluation

	IEEE	ACM	CSSL	SLASSCOM
5.1 What They Do Well	Global standards creation. 2,000+ conferences annually. Strong student programs worldwide. Leading technical knowledge source.	Comprehensive computing literature database. Open access from 2026. Specialized interest groups.	Bridges international standards with local context. Degree accreditation. NITC conference. GenZ student chapters.	G2C program addresses skills gap. Industry-education connection. Represents the industry to the government.
5.2 Problems and Weaknesses	Expensive professional membership (\$218/year).	Western-focused content	Less global recognition. Fewer online resources.	Benefits through employment only.
5.3 Usefulness for Sri Lankan Students	Best for academic careers and international opportunities.	Advanced research focus.	Highly relevant for local jobs.	Most valuable for employment.

6. Conclusion

Professional organizations are essential for ICT career development. IEEE and ACM provide global research and international recognition. CSSL maintains Sri Lankan professional standards and local networking. SLASSCOM connects graduates with employment opportunities.

7. References

ACM (2019). *About ACM*. [online] Acm.org. Available at: <https://www.acm.org/about-acm>.

Cssl.lk. (2024). *CSSL*. [online] Available at: <https://cssl.lk/about-us/>.

IEEE (2025). *About IEEE*. [online] Ieee.org. Available at: <https://www.ieee.org/about-ieee>.

Slasscom.lk. (2025). *About Us – SLASSCOM*. [online] Available at: <https://old.slasscom.lk/about-us/> [Accessed 4 Dec. 2025].